ICSIE 2026 · FULL PAPER PRESENTATION

SportSpace: Spatial Evidence for Planning Active Urban Environments in Ilidža–Stup

Semir Poturak · Amra Tuzović

University of Travnik, Faculty of Technical Sciences

Interactive pilot & paper: 3esign.github.io/sportspace-pilot

A map symbol is not an opportunity

Planning for an active city connects places to play and exercise with the routes and conditions through which they are used.

- WHO (2018, 2020): active environments are part of health-oriented urban planning
- But mapped proximity alone does not establish *usable access* or *health benefit*
- Early-stage question: which candidates merit examination, what does reaching them cost, what surrounds them, what could change the answer?

Three distinctions organise the study

1. Mapped candidate ≠ usable opportunity
2. Modelled opportunity ≠ actual participation
3. Health-relevant conditions ≠ demonstrated health effect

Question, contribution and position

> *How can modelled access, green context and historical environmental conditions be translated into transparent planning questions for sport and recreation, without treating them as verified participation or health outcomes?*

Proximity

Straight-line vs. network distance, per activity category

Green scale

Sensitivity of travel-time / vegetation trade-offs to buffer radius

Timing

Pollutant-specific comparison of historical departure times

Strand	Taken from it	Not claimed
Health-oriented planning (WHO 2016–2020)	Motivation to examine activity-supporting environments	Conversion from minutes or vegetation % to health effect
Accessibility measures (Geurs & van Wee 2004)	Nearest, cumulative, decay-weighted as complementary	Calibrated local preferences
Pollution-aware routing (Yan 2024) · Green & air (Venter 2024) · Vis design (Munzner 2009)	Trade-offs in travel; vegetation separate from air; interface as instrument	Street-level routing; vegetation as pollution reducer; proof the interface improves decisions

Contribution: an applied, reproducible workflow connecting each indicator to a decision and to the evidence required before acting on it.

Study design and data: Ilidža–Stup pilot

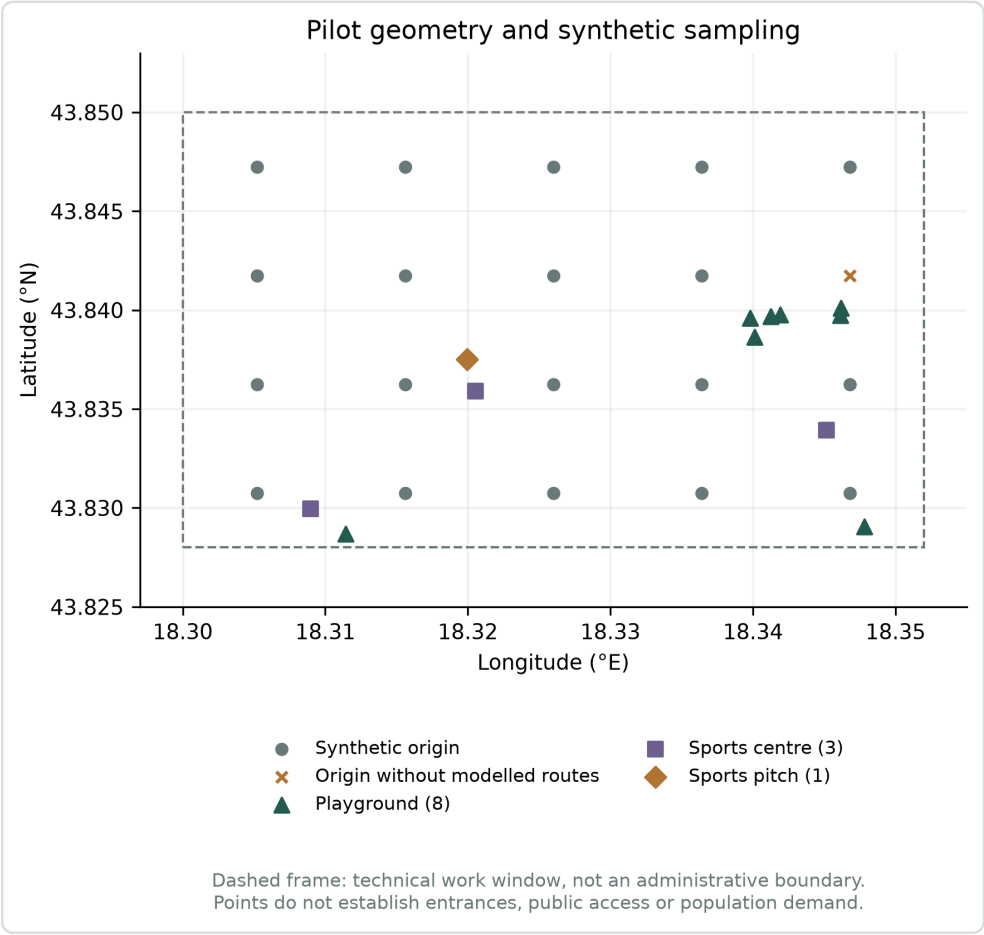
20 synthetic origins (4 × 5 grid cell centres) · **19** routable

12 mapped candidates: 8 playgrounds, 3 sports centres, 1 pitch

240 origin–destination pairs, **228** routable

Walking 1.2 m/s · 350 m snapping · OSM oneway treated as directed

Input	Support	Limitation
OpenStreetMap network + destinations	2026 snapshot	Entrances, access not field-verified
ESA WorldCover v200	2021, 10 m	No shade or public-access information
CAMS + ERA5 via Open-Meteo	8,760 h, 2021, one cell	Cannot distinguish neighbouring routes
Official Ilidža station pages	Documentary	No matched authorised hourly archive



Study geometry, baseline graph. Technical window 18.300–18.352° E, 43.828–43.850° N.

A 2026 network evaluated against 2021 environmental histories is a declared scenario, not a reconstruction of one observed year.

Method: three measures, one trade-off, one integral

Accessibility

Nearest destination: straight-line vs. network

Cumulative: $N_i(\tau) = \Sigma_j 1[t_{ij} \leq \tau]$, τ = 10 / 20 / 30 min

Potential: $P_i(h) = \Sigma_j 2^{-t_{ij}/h}$, h = 10 / 15 / 30 min

Green trade-off

$V_j(r)$: vegetation % at r = 100 / 300 / 500 m

Joint count with threshold g = 10 / 20 / 30 %

Pareto set: minimise t_{ij}, maximise $V_j(r)$

1,620 parameter rows; 57 radius comparisons per activity

Outing scenarios

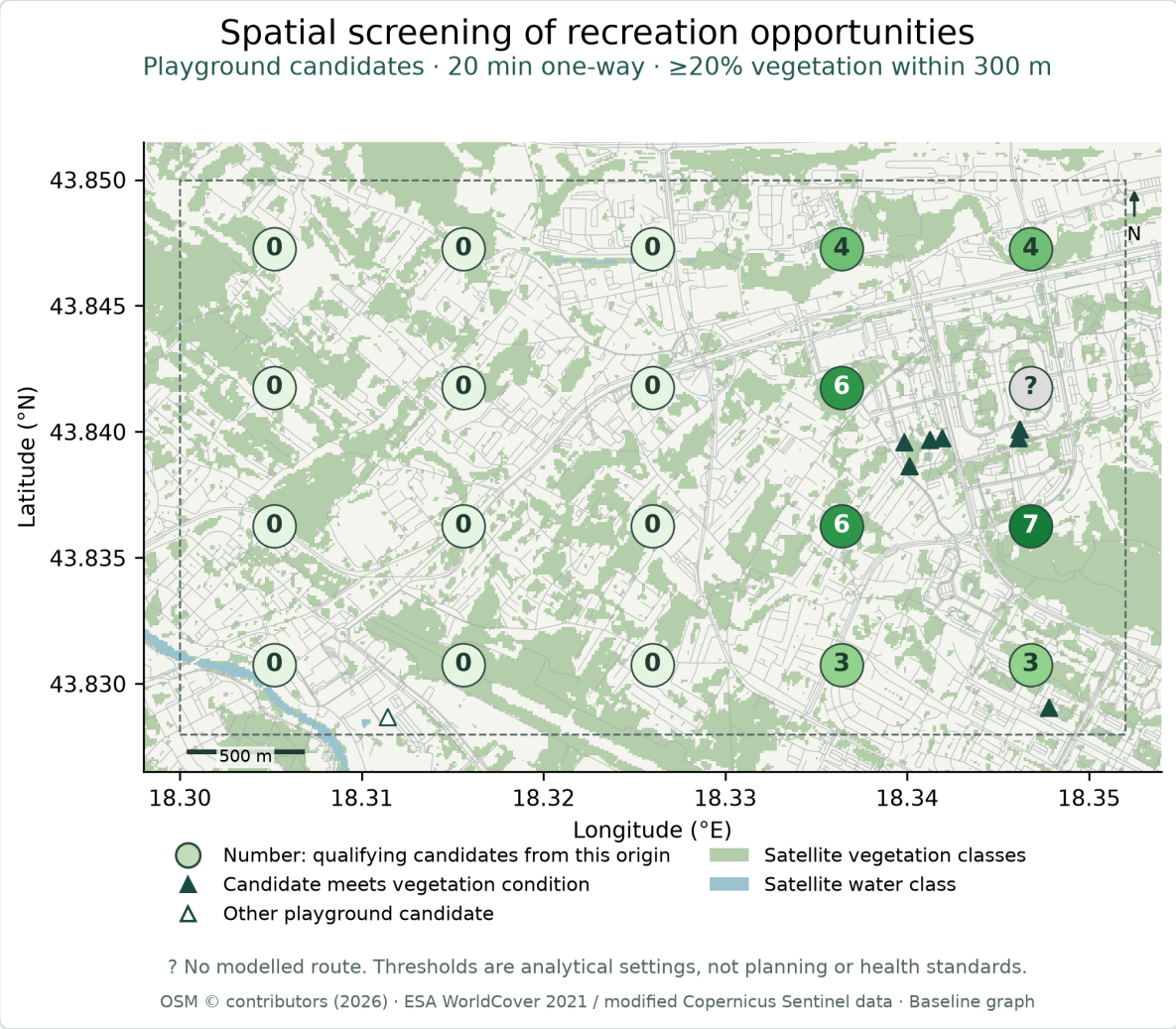
Departures 08:00 / 14:00 / 18:00, every 2021 date; stay 30 min (15, 60)

$E_p = \int_0^T C_p(s+u)\, du$, $T = 2t + d$

83,220 paired cases per pollutant; not inhaled dose

Equal destination weights. Thresholds are analytical scenarios, not health or planning standards.

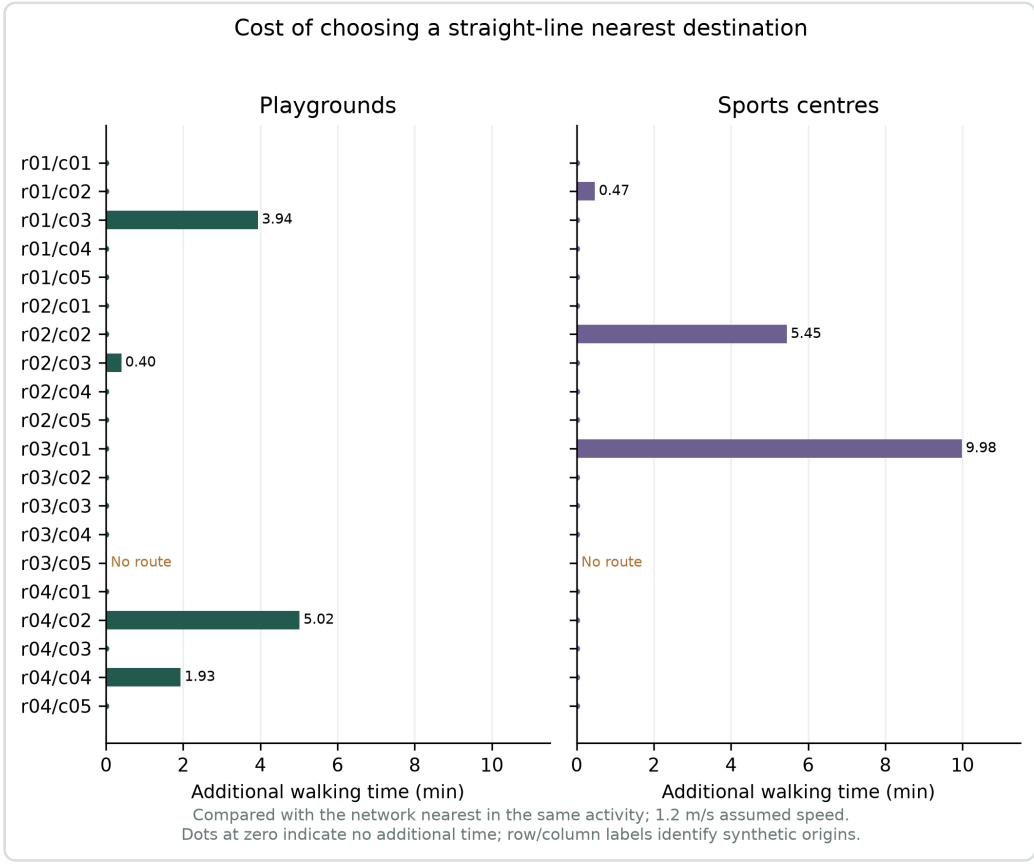
The screening scenario in space



Playground candidates within 20 assumed walking minutes with ≥20 % vegetation in 300 m. Counts are candidate records at synthetic origins, not catchments or population coverage.

Finding 1 · Network proximity changes the nearest destination

Activity	Dest.	Changed nearest	Max extra time
Playground	8	**4 / 19**	5.02 min
Sports centre	3	**3 / 19**	9.98 min
Sports pitch	1	0 / 19	0.00 min
All, descriptive	12	5 / 19	6.67 min



Extra modelled walking time when the straight-line nearest is chosen instead of the network nearest.

Conditional on baseline directionality, snapping and 1.2 m/s. Describes this synthetic origin set, not the travel burdens of residents.

Finding 2 · The scale of green context changes the compromise set

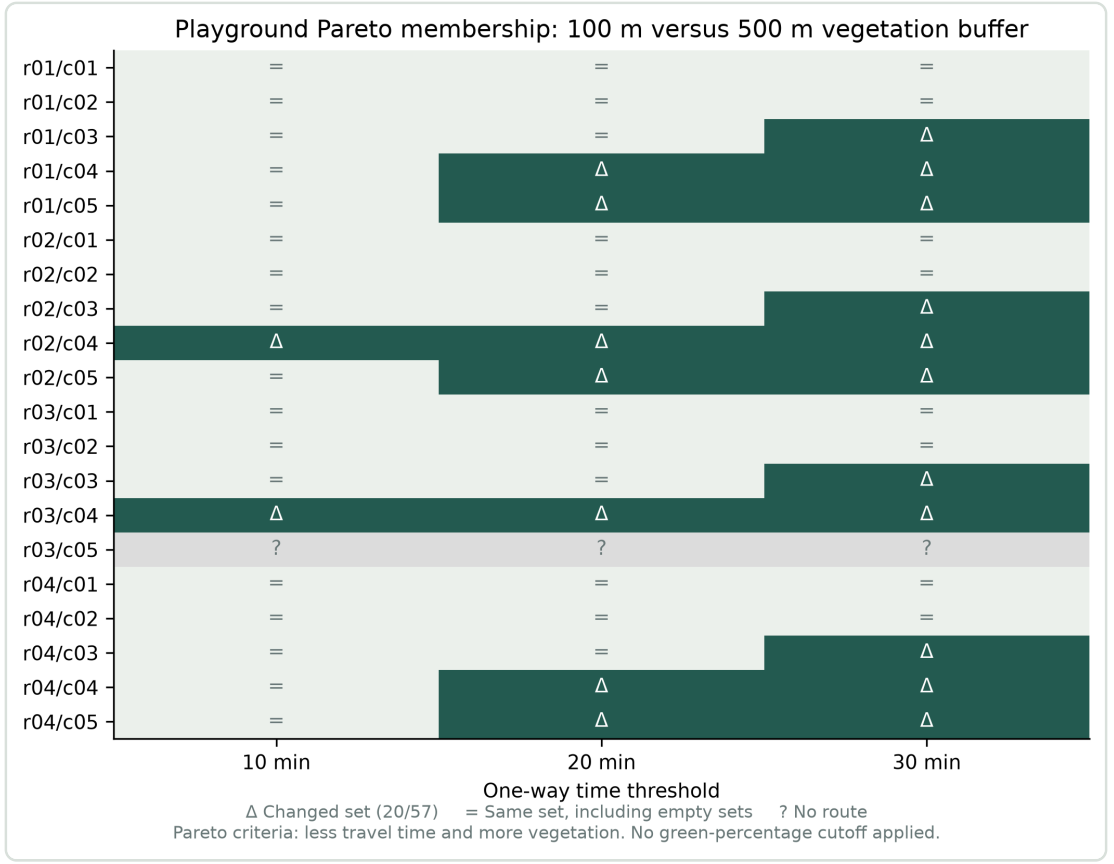
20 / 57 playground Pareto sets change between 100 m and 500 m buffers

0 / 57 for sports centres and for the single pitch

1,539 of 1,620 parameter rows have a routable origin; 81 remain unknown

Rounding vegetation to 0.001 pp: 3 / 513 sets change; to 1 pp: 17 / 513 (result becomes 19 / 57)

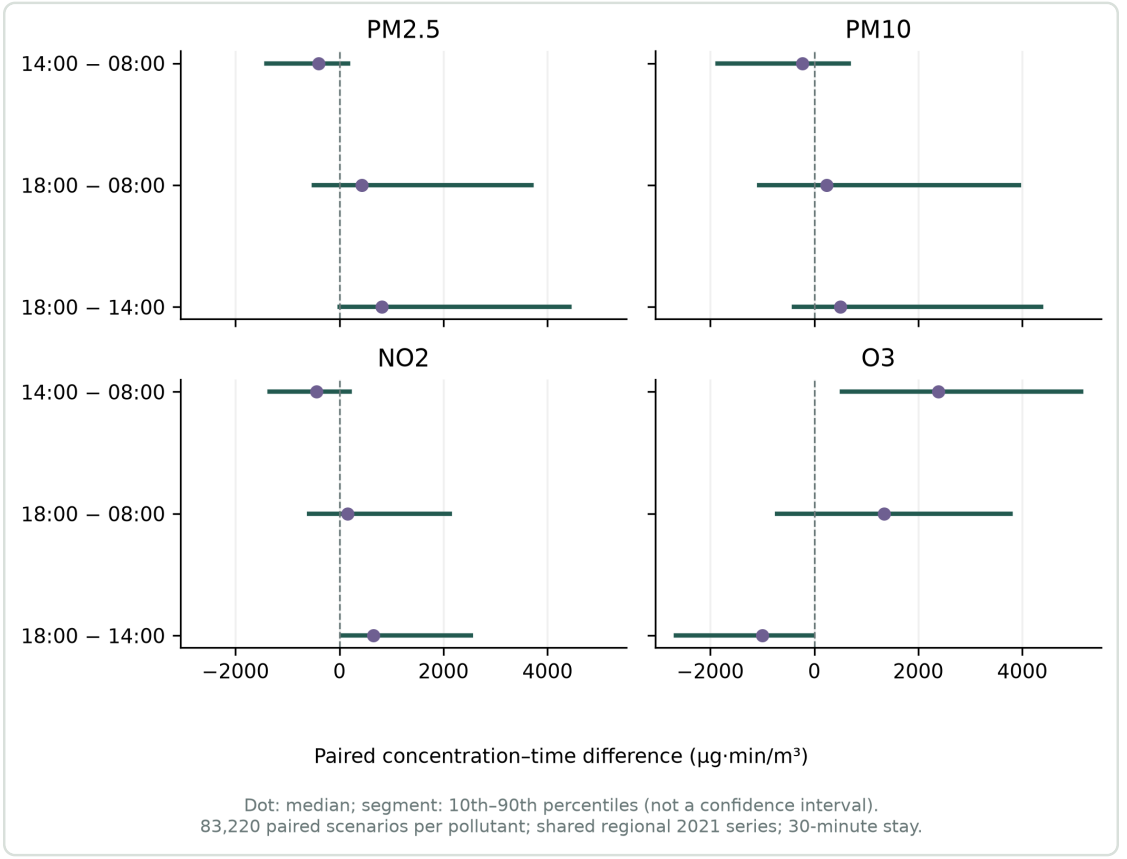
The broad finding persists; individual front membership can depend on small numerical differences. Not every difference is environmentally meaningful.



Pareto-set change per origin and time threshold.

Finding 3 · Departure-time comparisons differ between pollutants

Pollutant	Median $E_{14:00} - E_{08:00}$ (µg·min/m³)	Strict min at 14:00
PM2.5	−401.35	72.9 %
PM10	−224.75	44.8 %
NO2	−439.57	78.4 %
O3	**+2,389.75**	0.6 %



Medians and 10th–90th percentile ranges of 83,220 date–route scenarios per pollutant, 30-minute stay.

All cases share one regional 2021 series. Retrospective; does not recommend 14:00 today, nor a cleaner street.

Finding 4 · The representation itself is part of the result

Directionality

Removing graph direction rules shortens **125 / 228** routed pairs, max **11.45 min**. Sports-centre disagreement drops 3/19 → 2/19; playground set unchanged.

Candidate identity

Grouping the two unresolved close playground pairs changes opportunity counts in **15 / 57** origin–threshold cases (by at most two).

Connectors

Origin + destination connectors: median **2.13 %**, max **13.94 %** of baseline distance. All 20 origins satisfy 350 m snapping; 17 pass at 50 m, 19 at 100 m.

Independent rerouting reproduces all baseline distances. A record count should not be presented as an inventory of independent venues.

Controls and reproducibility

4,380 nested-interval monotonicity checks: no violations

80 origin–activity records and **1,620** rows verified; independent PM2.5 integration

10 core outputs regenerated twice offline, SHA-256 identical

Node.js standard library only, no npm dependencies

Station audit: official Ilidža displays exist, no matched authorised archive → no validation claim

Versioned analytical supplement with scripts, outputs, figures and manifests

Computational reproducibility, interpretive plausibility and real-world validity are three separate things. Tests and hashes establish only the first.

From indicator to planning question

Indicator	Concrete question	Permissible preliminary use	Does not establish
Network vs. straight-line; directional & connector sensitivity	Is the detour due to the city or the representation?	Review connections, crossings, entrances in records	A missing crossing or a justified construction project
Opportunity counts; unresolved grouping	How many distinct candidates remain?	Shortlist for documentary review (incl. physical education)	Capacity, suitability, permission, demand
Vegetation share; radius-dependent Pareto	Immediate setting or wider neighbourhood?	Examine spatial context of alternatives	Shade, usable green area, design quality
Pollutant-specific integrals; meteorology	Does the timing comparison change between pollutants?	Argue for temporal, pollutant-specific context	A safe time today or a cleaner route
Access documentation gaps	Which missing fact could invalidate the shortlist?	Assign a focused verification task	Access merely because an object is mapped

Example: the 9.98-minute sports-centre detour first suggests inspecting a missing link. The directionality audit shows the comparison changes with graph assumptions, so the first action is to inspect the representation, not to propose a physical connection.

Sport, physical education and health

Urban planning

Distribution and connection of potential activity settings

Urban design

Quality and continuity of route and place

Sport & PE practice

Can a specific activity be organised here for a specific group?

SportSpace supplies preliminary spatial information to this shared problem: which candidates to check, which cost more travel in the model, which details matter before organising activity.

Equipment, group size, supervision, surface, toilets, opening hours and permissions remain outside the analysed data. Health is upstream and conditional: opportunity ≠ participation ≠ benefit. The model cannot balance exercise benefit against pollution risk.

Limits and what comes next

Limits

- Synthetic origins: no population, schools, deprivation or demand
- 2026 network vs. 2021 environment: a mixed-year scenario
- Regional air and meteorology cannot differentiate nearby paths
- Pedestrian semantics, topology and connectors not field-verified
- Candidate identities and access incompletely verified
- No participants, activity minutes or health outcomes observed

Next, within the existing-data boundary

- Check pedestrian semantics and graph topology against source records
- Resolve close candidate identities; obtain documented venue properties
- Match authorised station archives if available
- Independently designed evaluation of decision usefulness

No CFD, new field campaigns or clinical research required for this contribution.

Conclusion

Network representation, the scale of vegetation context and the time of an assumed outing can each alter comparisons among mapped recreation candidates. Directionality and unresolved candidate identity do too.

The method organises the examination of **access, setting and timing** for sport and recreation, including the initial review of candidate settings for physical education. It cannot establish venue suitability, participation, equitable provision or health effects.

Defensible contribution: making the link from an indicator to a planning question and the next necessary evidence explicit, before a map-based result becomes a design or programme decision.

THANK YOU

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Live pilot, paper and supplement (BS / EN / TR):
3esign.github.io/sportspace-pilot

Data: OpenStreetMap © contributors · ESA WorldCover 2021 (modified Copernicus Sentinel data) · CAMS and ERA5 via Open-Meteo · FHMZ BiH